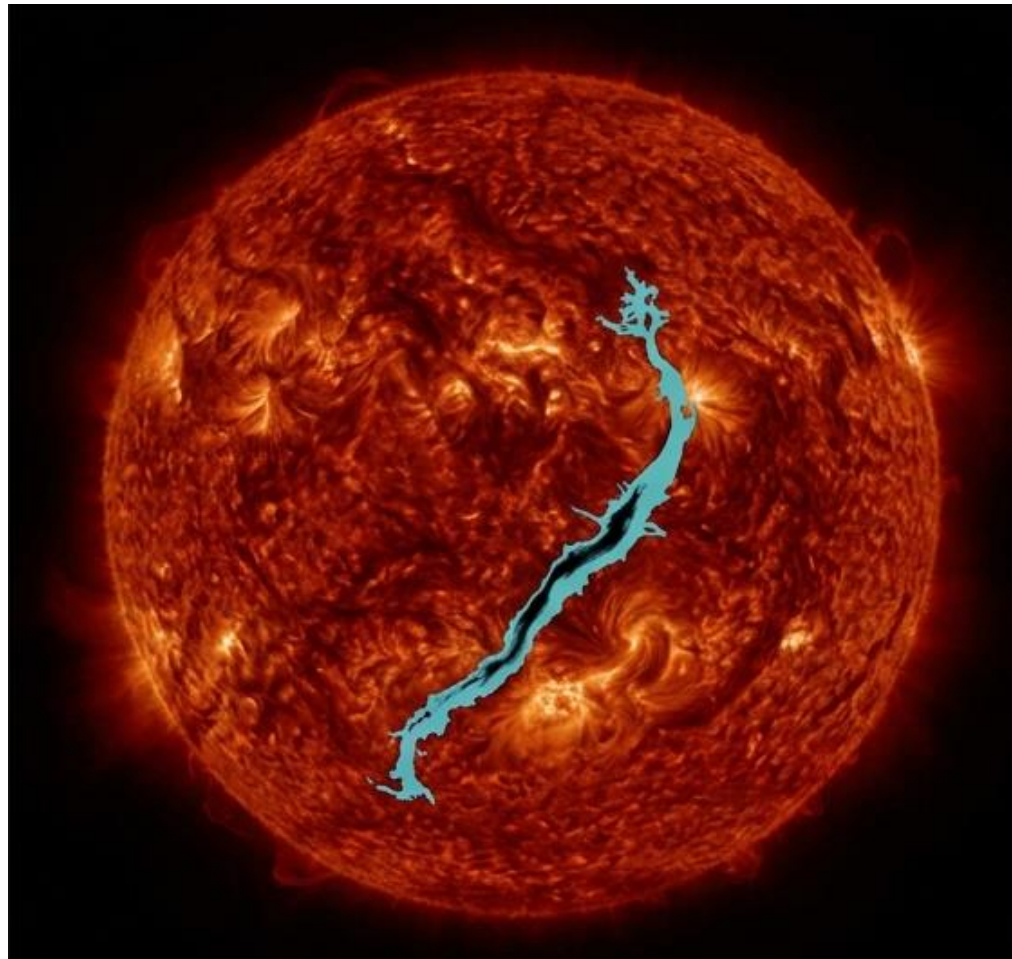


Machine Learning Project work: “Solar Filament segmentation”

Mario Vento and Diego Gragnaniello

Kaggle competition

- ◆ [Link to the training dataset](#)
- ◆ [Link to the competition description and rules](#)

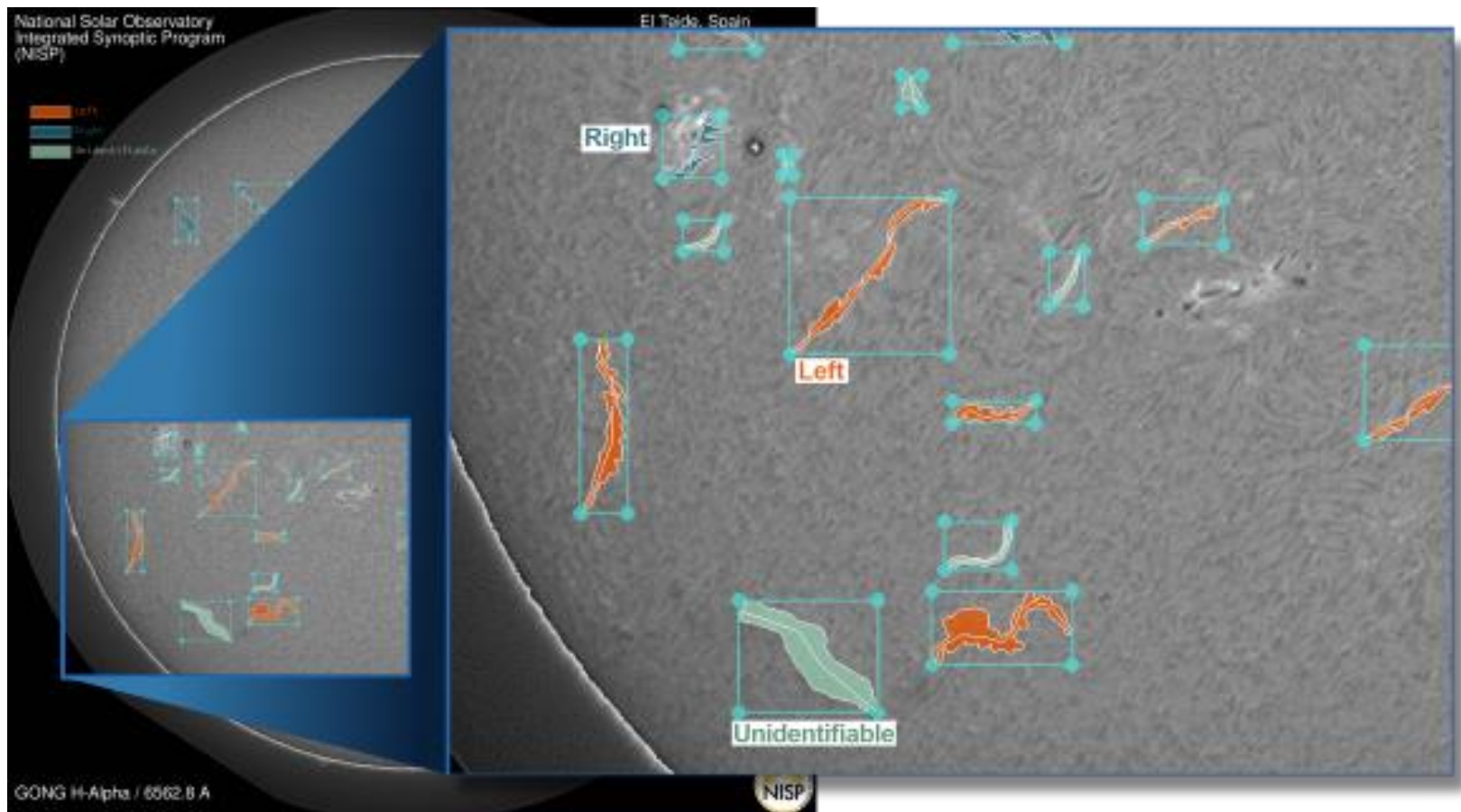


Kaggle competition

- ◆ [Link to the training dataset](#)
- ◆ [Link to the competition description and rules](#)
- ◆ You are required to:
 - ◆ Create your team on this [spreadsheet](#)
DEADLINE: August, 3rd at 10:00 AM
 - ◆ Create your team in Kaggle:
 - ◆ Using your account @studenti.unisa.it
 - ◆ The Kaggle team must match exactly the team on the spreadsheet
 - ◆ You cannot change the team after the deadline
 - ◆ You cannot make submissions in Kaggle with users that are not in your team

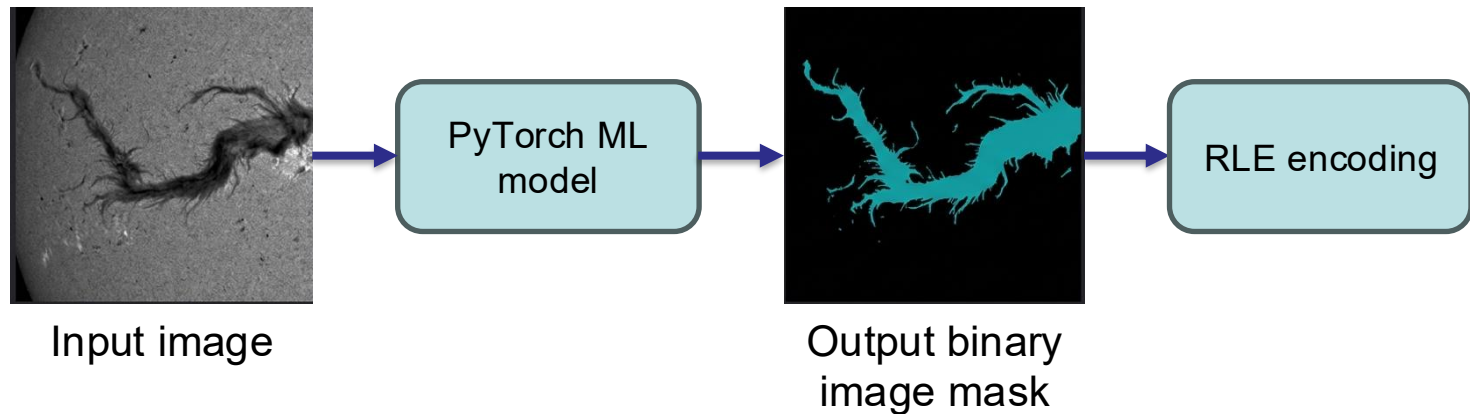
The task to address

- ◆ The goal is to segment filaments in solar images to produce a binary mask



The task to address

- ◆ The goal is to segment filaments in solar images to produce a binary mask
 - ◆ Method: any Machine Learning method that you can explain
 - ◆ Must be implemented in PyTorch and run in Colab
 - ◆ Submission in Kaggle after [converting the mask](#)



The task to address

- ◆ Challenges in the dataset:
 - ◆ Deal with background noise in the solar images
 - ◆ Produce segments that are not fragmented
 - ◆ Distinguish among filament and other background structures



Competition and rules

- ◆ Training images provided by the competition organizers
- ◆ You **cannot** extend the training set
- ◆ You ***can*** decide the training set split
- ◆ You ***can*** use data augmentation techniques
- ◆ You ***can*** create your own external test set

What you can use

- ◆ Any PyTorch algorithm (whatever kind of classifier, including non-neural ones, preprocessing, training strategy, validation), **BUT**:
 - You must be able to explain what you have used
- It must be **runnable with PyTorch**
 - inside **Google Colab and Kaggle**;
- Bonus (***but not mandatory***) if your solution uses:
 - less than **4 GB GPU RAM in test**
 - less than **5 GB GPU RAM in training**

IMPORTANT: Plan you experiments

- ◆ Do hypothesis on why your model is not working
 - ◆ Design an experiment to verify that specific hypothesis
 - ◆ Analyze results and comment them, thus select how to proceed
- ◆ Do a general plan of the experiments
 - Which hyperparameters you want to explore?
 - Which models/training paradigms to compare?
 - How much time does it takes?

IMPORTANT: Model training and validation is not a feed-forward process

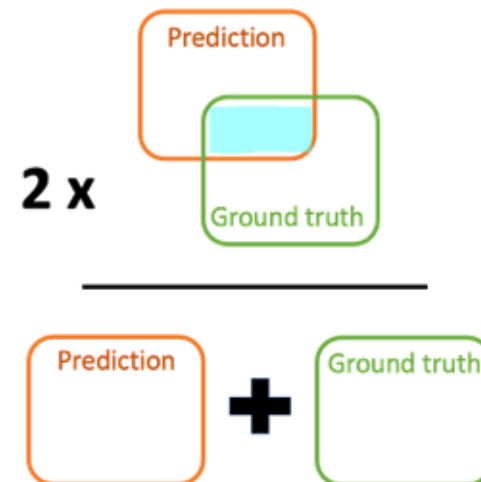
- ◆ Alternate trainings and validations:
 - ◆ After each validation, try to understand which samples are misclassified and why
 - ◆ Change your model and/or training set to improve the performance
 - ◆ If the performance seems very good, be sure that the validation set is challenging enough
- ◆ Keep track of your countermeasures (even if not successful) for the final project presentation

Model evaluation: **mandatory**

Performance will be evaluated:

- ◆ on the Kaggle **private test set** each time you **officially submit** your solution in Kaggle
 - Follow the [submission instructions](#)
- ◆ Submission in Kaggle **is mandatory** to evaluate your ML project work
- ◆ **Score adopted:** Mean Dice score computed in the Kaggle public leaderboard:

$$\text{Dice} = \frac{2 \times \text{Area of overlap}}{\text{Total area}} =$$



Model evaluation: **optional**

Note: Kaggle imposes additional rules and evaluations (see the competition website).

Those rules **are not mandatory** for the ML project work.

However, if you are compliant with all Kaggle rules, you may be eligible for Kaggle prizes at the end of the online competition.

Share the load

- ◆ Each member of the team will be requested to submit an estimate of the individual effort contributed by all members
 - To prevent "free riders"
 - Submissions will be "blind" (each member will not see the submissions of other members)

What you must submit

1. A link via Moodle to:
 1. a Google Drive folder
(**do not create a Shared Drive**)
 2. The folder must be renamed with your team name
(see [column B of the team composition spreadsheet](#))
 3. Make the link **accessible to all**
 4. **Do not add Professors** to directory members

What you must submit

2. The directory must contain:

- a) The Python Notebook to train the solution, eventually including the code to generate data using augmentation
- b) The train/validation split protocol (**NOT THE DATA**, but the sample list as CSV file)
- c) The files of the saved models/weights after the training
- d) The Python Notebook to test your system on the Kaggle test set and produce the submission file
- e) The full report (10 pages) and slides (5 minutes talk)

What you must submit: the presentation

- ◆ 5-minutes presentation (slide in English):
 1. Do not restate the problem/describe the dataset
 2. Go straight to your solution
 1. The rationale to collect/organize the dataset, including the dataset split into training and validation sets.
 2. The pre-processing pipelines adopted.
 3. The network architectures compared.
 4. The training hyperparameter setting, including the loss function.
 3. Let us know what challenges you encountered, how you tried to address them, what did not work and why, what countermeasure you adopted, what takehome messages you get from the results...

What you must submit: the report

◆ 10-pages font 12 report (in English):

1. Do not restate the problem/describe the dataset
2. Do not chronologically present your experiments without explaining how you plan them
3. Let us understand what hypothesis you did about the problem, how you managed to solve it and the results

If you want, you can present in Italian.

IMPORTANT: Don't forget to

- ◆ Write the **names of all the team members** in the Google Drive folder (in a text file)
- ◆ Ensure that the **link** you submit is **readable to anyone** (no authorization must be requested)
- ◆ Make sure that the **training and test scripts are readable**, commented and well organized

DEADLINE: beginning of July (TBD)

- ◆ Project work submission: **September, 4th at 4:00 PM**
 - Via Moodle
 - We consider the **Kaggle leaderboard freezed at this deadline**. No later submissions will be considered.
- ◆ Project work discussion: **September, 8th**
(time and room to be defined)
- ◆ All team members must discuss the project
(including those that book for their individual oral exam later)